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23BSIS-1P

Network Topologies

1. Mesh Topology

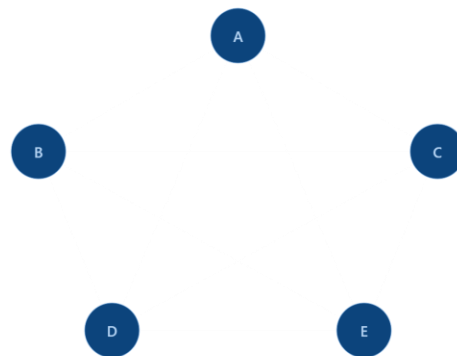
In a mesh topology, every device is connected directly to every other device in the network, creating multiple paths for data to travel.

Advantages:

- High fault tolerance — if one link fails, data can be rerouted through another path
- No traffic bottleneck since each connection has its own dedicated link
- Easy to identify faults since each link is independent

Disadvantages:

- Expensive to install and maintain due to the large amount of cabling required
- Difficult to scale (adding a new node requires many new connections)
- Complex configuration and management



Mesh Topology

2. Tree Topology

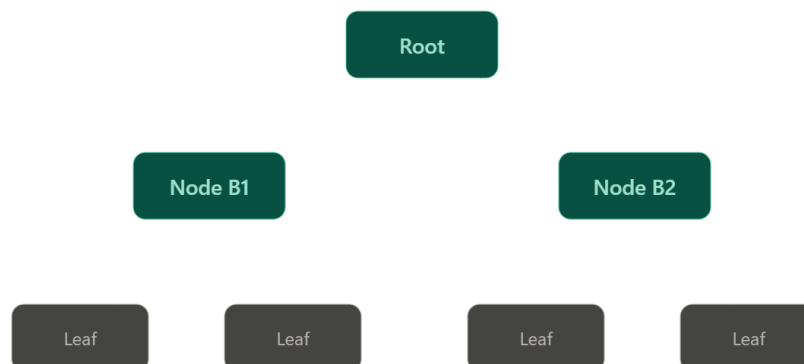
A tree topology combines characteristics of star and bus topologies, arranging nodes in a hierarchical, parent-child structure with a root node branching out to other nodes.

Advantages:

- Easy to scale by adding new branches without disturbing the rest of the network
- Fault isolation is simple since each branch can be managed independently
- Supports point-to-point wiring for individual segments, making troubleshooting easier

Disadvantages:

- If the root node or backbone fails, the entire network (or large segments) goes down
- Requires more cabling than simpler topologies like bus
- Maintenance becomes difficult as the network grows larger



Tree Topology

3. Ring Topology

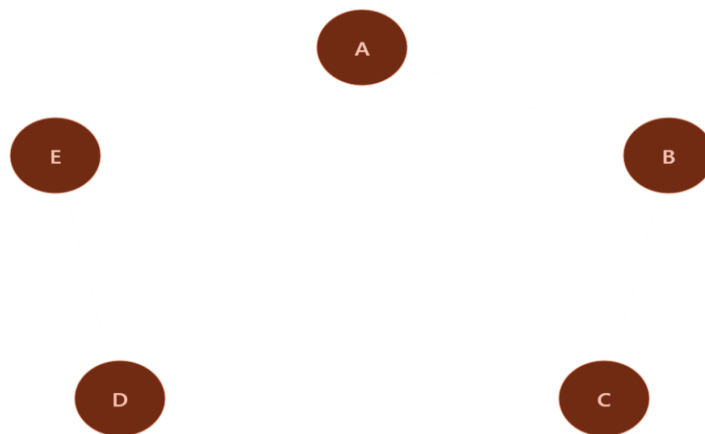
In a ring topology, each device connects to exactly two other devices, forming a circular data path where information travels in one (or both) direction(s) around the ring.

Advantages:

- Data transmission is fast and orderly, with no collisions since data flows in one direction
- Easier to install and manage compared to mesh
- Performs consistently well even under heavy network load

Disadvantages:

- A single point of failure in the ring can disrupt the entire network (unless it's a dual ring)
- Adding or removing devices can disrupt network operation
- Troubleshooting can be difficult since a fault anywhere affects the whole loop



Data travels in one direction around the loop